Winning in Major League Baseball & the Statistics that Drive It

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Introduction and Problem Statement

Business Problem

- We are functioning as a **data analytics team** working for a **Major League Baseball (MLB)** team's front office and given the following task by the team's general manager (GM): **Determine which team-level statistics contribute the most to increasing the team's season win total**

Context & Importance

- Our team's GM wants to have an edge when it comes to **scouting** for players to sign/draft in the offseason as well as pinpoint key areas of focus for **player development** in our team's farm system and believes that our analysis can **identify which key stats should be prioritized** when evaluating and coaching players in order to **increase team wins**

Objectives

- Our goals are to determine:
  1. Which **team statistics** have the greatest **impact on winning,** and
  2. To what extent

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Key Insights and Findings

Top Insights

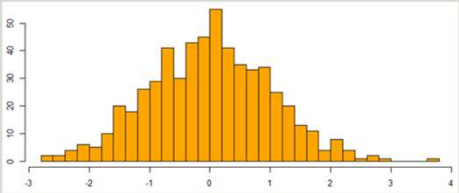
- Expected result: **runs created per 25 outs (RC_per25)** and **slugging percentage** have a **positive, significant** relationship with wins
- Unexpected result: **walks** and **sacrifice flies** have a **negative, significant** relationship with wins

Interpretation

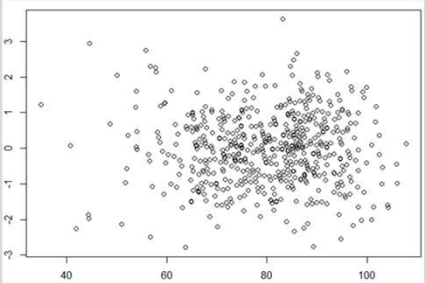
- Given all other predictors fixed:
  - A **1-unit increase** in RC_per25 is associated with **6.21 more wins**
  - A **0.010-unit increase** in slugging percentage is associated with **1.46 more wins**
  - A **1-unit increase** in walks is associated with **0.30 less wins**
  - A **1-unit increase** in sacrifice flies is associated with **0.27 less wins**

Visual Summaries

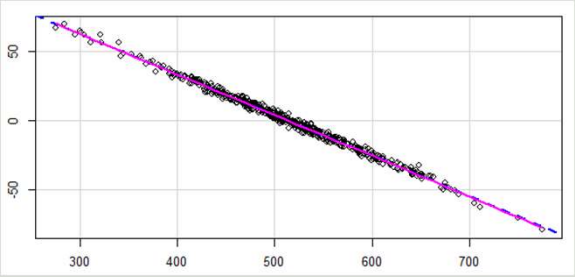
Distribution of Residuals



Fitted Values vs Residuals



Walks vs Residuals



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Business Impact and Implications

Implications of Findings

- Every season we want to **win as many games as possible**, and our model shows us that **high levels of runs created and slugging percentages** are some of the more impactful stats that will help us capture those wins
- A contributing factor to these stats is **advancing around the bases**, through hits and general baserunning

Strategic Value

- Our analysis has given us specific player traits we'll be looking for (extra base hitting, baserunning) when it comes to **scouting, trading, and signing players**
- Knowing exactly what we are looking for will give us an advantage in negotiating contracts with players; we will be able to **avoid bidding wars over the most popular players** if they do not fit our goals and we will also be able to **negotiate good deals on under-the-radar players**

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Recommendations

Actionable Recommendations

- Build a team identity and set a culture: we want a **smart, speedy team** that will be able to punish defenses so that they can **advance around the bases**
- Ensure everybody in the organization is bought into the plan by emphasizing the vision to the key team decision-makers and at all levels of the organization

Implementation Steps

- **Evaluate roster** to determine if/how current players fit into new team identity
- **Sign, draft, or trade for players** that will be able to immediately contribute to new goals
- **Build a player development system** that will grow players in the minor leagues into what we need out of them if/when they get promoted to the MLB roster

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Implementation Plan

High-Level Plan

- Use model to **rank players by predicted value**
- Sign, draft, and trade for players that provide the **highest value**
- Continuous **data collection** and **processing**
- **Continuous improvement of model** and expansion of analytics strategy

Resources and Support

Resources

- **Data storage**
- **Visualization tools** to display model results
- **Funding** to expand analytics team

Support

- **Buy-in** from management, scouts, and coaches

Timeline

Month 1	Introduce model to scouts and coaches
Month 2-3	Integrate model into player evaluation approach
Month 4-5	Conduct training sessions
Month 6-7	Deploy model across entire team and use in decision-making process
On going	Continuous improvement of model

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Challenges and Considerations

Potential Challenges	Mitigation Strategies
• Resistance from scouts and coaches to adopt the model	• Educate team and involve scouts and coaches throughout development process
• Overfitting: the model may perform well on historical data but fail to generalize new observations	• Validate and test on multiple datasets to ensure accuracy of model
• Changing model results as new data is introduced	• Be cognizant of changes and frequently rerun model with new data

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Future Directions

Further Analysis

- During our model evaluation, we identified several variables that had **unexpectedly negative coefficients in relation to wins**, despite initial expectations that they would be positive
- We suspect that these unexpectedly negative coefficients could be the result of **our model not capturing all linear relationships impacting wins**, so further analysis is needed to determine which additional variables may be needed (i.e., in-game strategies)

Variable	Coefficient	P-value
Walks	-0.30	p < 2e-16***
Sacrifice Flies	-0.28	p < 2e-16***

Long-Term Improvement

1. **Data Enrichment**
   - Contextual dependencies, where the real-world scenarios affect the variables differently than expected
   - Integrate additional data sources, such as in-game strategies, and psychological factors, to enrich the model
2. **Dynamic Model Update**
   - Implement a dynamic model that updates with new season data to improve accuracy and relevance

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Conclusion and Summary

Recap of Key Points

- Verified correlations between variables and wins through **exploratory data analysis**
- Identified **14** independent variables are statistically significant at **95%** out of **42** using **linear regression** and **predictive modeling**
- **Validated five models** and selected the most suitable model by considering the RMSE and adjusted R-squared values
- From this model, we found that high levels of **runs created** and **slugging percentages significantly contribute to more wins**
- On the other hand, **walks** and **sacrifice flies** showed a **negative and significant relationship** with wins, contrary to expectation

Concluding Thoughts

- Using the data analysis methods learned this semester, we performed raw data analysis and cleaning, and **selected the optimal model** to meet our objectives
- For the variables of unexpected results, we plan to **enhance the model** by **adding data** from the new season and real-world scenarios. We'll **apply a dynamic model** for fine-tuning and further analysis to improve accuracy
- As specified in the timeline, we will **continuously improve the model** so that it can be utilized in team operations within a month

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